

General Algebraic Modeling System
Compilation

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1  * =====
2  * Telecom Staff Scheduling (GAMS) - all 3 notebook parts
3  * =====
4
5  Sets
6      h   "hours"      /h0*h23/
7      j   "companies" /c1*c2/
8      peak(h) "hours 7..17 (inclusive)"
9      off(h)  "hours 0..6 and 18..23"
10     Q(h,h)  "coverage mapping: for each demand hour, staffed start hours in
6-hour window";
11
12     Alias (h,i,k);
13
14     * Subsets for objective weights (ord(h0)=1)
15     peak(h) = yes$(ord(h) >= 8 and ord(h) <= 18);
16     off(h)  = yes$(ord(h) <= 7 or ord(h) >= 19);
17
18     Parameters
19         cost(h)          "cost per operator starting at hour"
20         demand(h,j)      "demand by hour and company"
21         capHour(h)        "RHS for per-hour starting cap"
22         capTotal(h)       "RHS for total active cap per demand hour"
23         minCov(h)         "minimum coverage per demand hour"
24         maxShort(h,j)     "max shortage per hour/company"
25         totShort(j)       "total shortage cap by company";
26
27     * --- Default RHS values (so we can vary them later in part 3)
28     capHour(h) = 100;
29     capTotal(h)= 420;
30     minCov(h)  = 80;
31     maxShort(h,j) = 10;
32     totShort('c1') = 40;
33     totShort('c2') = 30;
34
35     * Costs (hour 0..23)
36     cost('h0') = 189; cost('h1') = 189; cost('h2') = 189; cost('h3') = 180;
37     cost('h4') = 171; cost('h5') = 162; cost('h6') = 153; cost('h7') = 144;
38     cost('h8') = 135; cost('h9') = 126; cost('h10') = 126; cost('h11') = 135;
39     cost('h12') = 144; cost('h13') = 153; cost('h14') = 162; cost('h15') = 171;
40     cost('h16') = 180; cost('h17') = 189; cost('h18') = 189; cost('h19') = 189;
41     cost('h20') = 189; cost('h21') = 189; cost('h22') = 189; cost('h23') = 189;
42
43     * Demand (hour, company)

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44 * Company 1
45 demand('h0','c1') = 120; demand('h1','c1') = 95; demand('h2','c1') = 90;
demand('h3','c1') = 50;
46 demand('h4','c1') = 30; demand('h5','c1') = 30; demand('h6','c1') = 60;
demand('h7','c1') = 110;
47 demand('h8','c1') = 140; demand('h9','c1') = 200; demand('h10','c1') = 180;
demand('h11','c1') = 180;
48 demand('h12','c1') = 240; demand('h13','c1') = 220; demand('h14','c1') = 220;
demand('h15','c1') = 230;
49 demand('h16','c1') = 250; demand('h17','c1') = 260; demand('h18','c1') = 230;
demand('h19','c1') = 250;
50 demand('h20','c1') = 260; demand('h21','c1') = 250; demand('h22','c1') = 220;
demand('h23','c1') = 190;
51
52 * Company 2
53 demand('h0','c2') = 70; demand('h1','c2') = 40; demand('h2','c2') = 20;
demand('h3','c2') = 30;
54 demand('h4','c2') = 50; demand('h5','c2') = 40; demand('h6','c2') = 40;
demand('h7','c2') = 90;
55 demand('h8','c2') = 110; demand('h9','c2') = 110; demand('h10','c2') = 110;
demand('h11','c2') = 130;
56 demand('h12','c2') = 150; demand('h13','c2') = 160; demand('h14','c2') = 120;
demand('h15','c2') = 130;
57 demand('h16','c2') = 160; demand('h17','c2') = 180; demand('h18','c2') = 200;
demand('h19','c2') = 170;
58 demand('h20','c2') = 140; demand('h21','c2') = 110; demand('h22','c2') = 100;
demand('h23','c2') = 90;
59
60 * Coverage mapping Q(i,k): 6-hour cyclic window (wraps around midnight)
61 Q('h0','h18')=yes; Q('h0','h19')=yes; Q('h0','h20')=yes; Q('h0','h22')=yes;
Q('h0','h23')=yes; Q('h0','h0')=yes;
62 Q('h1','h19')=yes; Q('h1','h20')=yes; Q('h1','h21')=yes; Q('h1','h23')=yes;
Q('h1','h0')=yes; Q('h1','h1')=yes;
63 Q('h2','h20')=yes; Q('h2','h21')=yes; Q('h2','h22')=yes; Q('h2','h0')=yes;
Q('h2','h1')=yes; Q('h2','h2')=yes;
64 Q('h3','h21')=yes; Q('h3','h22')=yes; Q('h3','h23')=yes; Q('h3','h1')=yes;
Q('h3','h2')=yes; Q('h3','h3')=yes;
65 Q('h4','h22')=yes; Q('h4','h23')=yes; Q('h4','h0')=yes; Q('h4','h2')=yes;
Q('h4','h3')=yes; Q('h4','h4')=yes;
66 Q('h5','h23')=yes; Q('h5','h0')=yes; Q('h5','h1')=yes; Q('h5','h3')=yes;
Q('h5','h4')=yes; Q('h5','h5')=yes;
67 Q('h6','h0')=yes; Q('h6','h1')=yes; Q('h6','h2')=yes; Q('h6','h4')=yes;
Q('h6','h5')=yes; Q('h6','h6')=yes;
68 Q('h7','h1')=yes; Q('h7','h2')=yes; Q('h7','h3')=yes; Q('h7','h5')=yes;
Q('h7','h6')=yes; Q('h7','h7')=yes;
69 Q('h8','h2')=yes; Q('h8','h3')=yes; Q('h8','h4')=yes; Q('h8','h6')=yes;
Q('h8','h7')=yes; Q('h8','h8')=yes;
70 Q('h9','h3')=yes; Q('h9','h4')=yes; Q('h9','h5')=yes; Q('h9','h7')=yes;
Q('h9','h8')=yes; Q('h9','h9')=yes;
71 Q('h10','h4')=yes; Q('h10','h5')=yes; Q('h10','h6')=yes; Q('h10','h8')=yes;

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Q('h10','h9')=yes; Q('h10','h10')=yes;
72  Q('h11','h5')=yes; Q('h11','h6')=yes; Q('h11','h7')=yes; Q('h11','h9')=yes;
Q('h11','h10')=yes;Q('h11','h11')=yes;
73  Q('h12','h6')=yes; Q('h12','h7')=yes; Q('h12','h8')=yes;
Q('h12','h10')=yes;Q('h12','h11')=yes;Q('h12','h12')=yes;
74  Q('h13','h7')=yes; Q('h13','h8')=yes; Q('h13','h9')=yes;
Q('h13','h11')=yes;Q('h13','h12')=yes;Q('h13','h13')=yes;
75  Q('h14','h8')=yes; Q('h14','h9')=yes;
Q('h14','h10')=yes;Q('h14','h12')=yes;Q('h14','h13')=yes;Q('h14','h14')=yes;
76  Q('h15','h9')=yes;
Q('h15','h10')=yes;Q('h15','h11')=yes;Q('h15','h13')=yes;Q('h15','h14')=yes;Q('h15',
'h15')=yes;
77
Q('h16','h10')=yes;Q('h16','h11')=yes;Q('h16','h12')=yes;Q('h16','h14')=yes;Q('h16',
'h15')=yes;Q('h16','h16')=yes;
78
Q('h17','h11')=yes;Q('h17','h12')=yes;Q('h17','h13')=yes;Q('h17','h15')=yes;Q('h17',
'h16')=yes;Q('h17','h17')=yes;
79
Q('h18','h12')=yes;Q('h18','h13')=yes;Q('h18','h14')=yes;Q('h18','h16')=yes;Q('h18',
'h17')=yes;Q('h18','h18')=yes;
80
Q('h19','h13')=yes;Q('h19','h14')=yes;Q('h19','h15')=yes;Q('h19','h17')=yes;Q('h19',
'h18')=yes;Q('h19','h19')=yes;
81
Q('h20','h14')=yes;Q('h20','h15')=yes;Q('h20','h16')=yes;Q('h20','h18')=yes;Q('h20',
'h19')=yes;Q('h20','h20')=yes;
82
Q('h21','h15')=yes;Q('h21','h16')=yes;Q('h21','h17')=yes;Q('h21','h19')=yes;Q('h21',
'h20')=yes;Q('h21','h21')=yes;
83
Q('h22','h16')=yes;Q('h22','h17')=yes;Q('h22','h18')=yes;Q('h22','h20')=yes;Q('h22',
'h21')=yes;Q('h22','h22')=yes;
84
Q('h23','h17')=yes;Q('h23','h18')=yes;Q('h23','h19')=yes;Q('h23','h21')=yes;Q('h23',
'h22')=yes;Q('h23','h23')=yes;
85
86
87  * =====
88  * PART 1: Base MIP
89  * =====
90
91  Variables
92      z "total cost (objective)";
93
94  Integer Variables
95      x(h,j) "operators starting at hour for company"
96      L(h,j) "shortage at hour for company";
97
98  Equations

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99      obj          "objective definition"
100     cap_per_hour(h)  "sum of operators starting at hour across companies <=
capHour"
101     cap_total(h)     "total operators active in window across companies <=
capTotal"
102     shortage(h,j)    "demand - covered <= L"
103     shortage_cap(h,j) "L <= maxShort"
104     shortage_total(j) "sum_h L(h,j) <= totShort(j)"
105     min_coverage(h)   "minimum coverage in each window across companies >=
minCov";
106
107  obj..
108      z =e=
109          sum((h,j), cost(h)*x(h,j))
110          + 60*sum(i$peak(i), L(i,'c1'))
111          + 30*sum(i$off(i), L(i,'c1'))
112          + 45*sum(i, L(i,'c2'));
113
114  cap_per_hour(h)..
115      sum(j, x(h,j)) =l= capHour(h);
116
117  cap_total(h)..
118      sum((k,j)$Q(h,k), x(k,j)) =l= capTotal(h);
119
120  shortage(h,j)..
121      demand(h,j) - sum(k$Q(h,k), x(k,j)) =l= L(h,j);
122
123  shortage_cap(h,j)..
124      L(h,j) =l= maxShort(h,j);
125
126  shortage_total(j)..
127      sum(h, L(h,j)) =l= totShort(j);
128
129  min_coverage(h)..
130      sum((k,j)$Q(h,k), x(k,j)) =g= minCov(h);
131
132  Model Telecom_MIP /all/;
133
134  Solve Telecom_MIP using MIP minimizing z;
135
136  display "PART 1 (MIP) objective", z.l;

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COMPILATION TIME      =          0.000 SECONDS          3 MB  52.5.0 9bc771bf WEX-WEI
GAMS 52.5.0 9bc771bf Jan 28, 2026          WEX-WEI x86 64bit/MS Windows - 02/15/26
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General Algebraic Modeling System
Equation Listing      SOLVE Telecom_MIP Using MIP From line 134

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---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 0)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;
(LHS = 0)

cap_total(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS
= 0)

cap_total(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS
= 0)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = 0, INFES = 120 ****)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = 0, INFES = 70 ****)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = 0, INFES = 95 ****)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 0)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 0)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 0, INFES = 80 ****)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 0, INFES = 80 ****)

```

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 0, INFES = 80 ****)

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REMAINING 21 ENTRIES SKIPPED

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Column Listing SOLVE Telecom_MIP Using MIP From line 134

---- z total cost (objective)

z

```

          (.LO, .L, .UP, .M = -INF, 0, +INF, 0)
1          obj

```

---- x operators starting at hour for company

x(h0,c1)

```

          (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-189      obj
1          cap_per_hour(h0)
1          cap_total(h0)
1          cap_total(h1)
1          cap_total(h2)
1          cap_total(h4)
1          cap_total(h5)
1          cap_total(h6)
-1          shortage(h0,c1)
-1          shortage(h1,c1)
-1          shortage(h2,c1)
-1          shortage(h4,c1)
-1          shortage(h5,c1)
-1          shortage(h6,c1)
1          min_coverage(h0)
1          min_coverage(h1)
1          min_coverage(h2)
1          min_coverage(h4)
1          min_coverage(h5)
1          min_coverage(h6)

```

x(h0,c2)

```

          (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-189      obj
1          cap_per_hour(h0)
1          cap_total(h0)
1          cap_total(h1)

```

```

1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

```

x(h1,c1)
      (.L0, .L, .UP, .M = 0, 0, +INF, 0)
-189  obj
1      cap_per_hour(h1)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)
1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.L0, .L, .UP, .M = 0, 0, +INF, 0)
-30   obj
-1     shortage(h0,c1)
1      shortage_cap(h0,c1)
1      shortage_total(c1)

```



```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 134

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 134

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 134

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	134

```

**** SOLVER STATUS      1 Normal Completion
**** MODEL STATUS      1 Optimal
**** OBJECTIVE VALUE    184848.0000

```

```

RESOURCE USAGE, LIMIT    0.062 10000000000.000
ITERATION COUNT, LIMIT   82    2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

```

```

--- MIP status (101): integer optimal solution.
--- Cplex Time: 0.06sec (det. 12.53 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

```

Proven optimal solution
MIP Solution:      184848.000000    (82 iterations, 0 nodes)
Final Solve:      184848.000000    (0 iterations)

Best possible:     184848.000000
Absolute gap:      0.000000
Relative gap:      0.000000

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour  sum of operators starting at hour across companies <= capHour

```

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.

h1	-INF	5.0000	100.0000	.
h2	-INF	28.0000	100.0000	.
h3	-INF	13.0000	100.0000	.
h4	-INF	32.0000	100.0000	.
h5	-INF	30.0000	100.0000	.
h6	-INF	24.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	53.0000	100.0000	.
h9	-INF	82.0000	100.0000	.
h10	-INF	70.0000	100.0000	.
h11	-INF	43.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	37.0000	100.0000	.
h14	-INF	43.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	55.0000	100.0000	.
h17	-INF	85.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	50.0000	100.0000	.
h20	-INF	42.0000	100.0000	.
h21	-INF	28.0000	100.0000	.
h22	-INF	7.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	199.0000	420.0000	.
h1	-INF	125.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	81.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	119.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	291.0000	420.0000	.
h11	-INF	349.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	415.0000	420.0000	.
h14	-INF	385.0000	420.0000	.
h15	-INF	375.0000	420.0000	.
h16	-INF	411.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	415.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.

h22	-INF	317.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-129.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-51.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-37.0000	-30.0000	.
h5 .c2	-INF	-43.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-59.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-181.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-219.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-255.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-165.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-145.0000	-130.0000	.
h16.c1	-INF	-251.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.

h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	3.0000	10.0000	.
h1 .c2	-INF	7.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	-1.522184E-8	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	4.0000	10.0000	.
h18.c2	-INF	6.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	5.0000	10.0000	.

h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	3.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	199.0000	+INF	.
h1	80.0000	125.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	81.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	119.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	291.0000	+INF	.
h11	80.0000	349.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	415.0000	+INF	.
h14	80.0000	385.0000	+INF	.
h15	80.0000	375.0000	+INF	.
h16	80.0000	411.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	415.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	317.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184848.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	5.0000	+INF	189.0000
h2 .c1	.	23.0000	+INF	189.0000
h2 .c2	.	5.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	13.0000	+INF	180.0000
h4 .c1	.	7.0000	+INF	171.0000
h4 .c2	.	25.0000	+INF	171.0000
h5 .c1	.	30.0000	+INF	162.0000
h5 .c2	.	-2.410219E-8	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	24.0000	+INF	153.0000
h7 .c1	.	57.0000	+INF	144.0000
h7 .c2	.	43.0000	+INF	144.0000
h8 .c1	.	53.0000	+INF	135.0000
h8 .c2	.	-3.932403E-8	+INF	135.0000
h9 .c1	.	53.0000	+INF	126.0000
h9 .c2	.	29.0000	+INF	126.0000
h10.c1	.	38.0000	+INF	126.0000
h10.c2	.	32.0000	+INF	126.0000
h11.c1	.	41.0000	+INF	135.0000
h11.c2	.	2.0000	+INF	135.0000
h12.c1	.	51.0000	+INF	144.0000
h12.c2	.	49.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	37.0000	+INF	153.0000
h14.c1	.	25.0000	+INF	162.0000
h14.c2	.	18.0000	+INF	162.0000
h15.c1	.	73.0000	+INF	171.0000
h15.c2	.	27.0000	+INF	171.0000
h16.c1	.	23.0000	+INF	180.0000
h16.c2	.	32.0000	+INF	180.0000
h17.c1	.	62.0000	+INF	189.0000
h17.c2	.	23.0000	+INF	189.0000
h18.c1	.	65.0000	+INF	189.0000
h18.c2	.	35.0000	+INF	189.0000
h19.c1	.	25.0000	+INF	189.0000
h19.c2	.	25.0000	+INF	189.0000
h20.c1	.	39.0000	+INF	189.0000
h20.c2	.	3.0000	+INF	189.0000
h21.c1	.	28.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000

h22.c1	.	-3.180922E-9	+INF	189.0000
h22.c2	.	7.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	3.0000	+INF	30.0000
h1 .c2	.	7.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	-1.522184E-8	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	4.0000	+INF	30.0000
h18.c2	.	6.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	5.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000

h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	3.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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 General Algebraic Modeling System
 Execution

---- 136 PART 1 (MIP) objective
 VARIABLE z.L = 184848.000 total cost (objective)

EXECUTION TIME = 0.188 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI

USER: GAMS Demo, for EULA and demo limitations see G251112+0003Cc-GEN
 <https://www.gams.com/latest/docs/UG%5FLicense.html> DC0000

**** FILE SUMMARY

Input C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\telecom_part1.gms
 Output C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\telecom_part1.lst